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Scenario – 62

SE LAB - 01

Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Donor Partners to list surplus food batches with dietary tags, total quantity, and a perishable expiry countdown timer.	High	Pass: A new food batch is visible on the platform with accurate tags and an active countdown timer. Fail: A batch is published without a mandatory expiry time.	Establishes the core inventory of the platform and prevents the distribution of unsafe food.
FR-002	Functional	The system shall automatically unlist and flag food batches	High	Pass: Expired batch immediately disappears from the active claim	Ensures strict compliance with food safety and

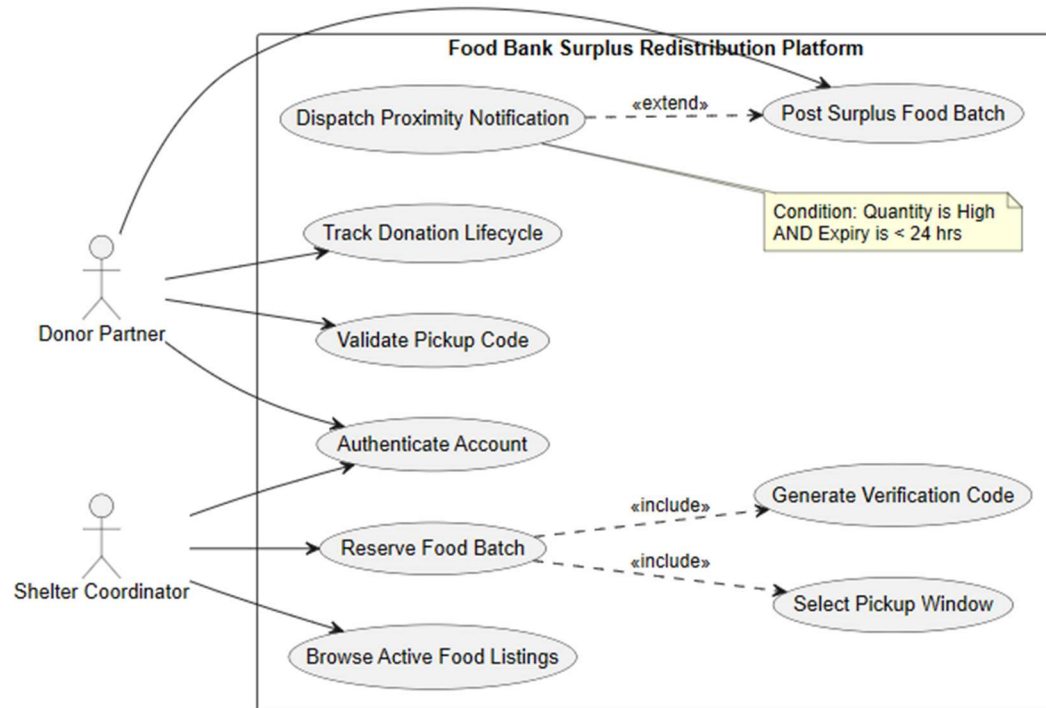
ID	Type	Description	Priority	Acceptance Criteria	Rationale
		as unavailable once their perishable expiry countdown timer reaches zero.		board and cannot be reserved. Fail: Expired food batch remains active and claimable by shelters.	health regulations.
FR-003	Functional	The system shall enable Shelter Coordinators to reserve available food batches and select a designated pickup time window.	High	Pass: Inventory quantity decreases or is marked "Reserved" upon successful selection of a pickup window. Fail: Multiple shelters can reserve the exact same batch simultaneously.	Facilitates organized redistribution and prevents double-booking of resources.
FR-004	Functional	The system shall generate a secure,	Medium	Pass: Status updates to "Picked Up"	Ensures food is securely handed over

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		unique alphanumeric pickup verification code for the Shelter Coordinator upon reservation, which the Donor Partner must validate at handover.		only after the Donor Partner enters the matching code. Fail: Handover completes without code validation.	to the verified and intended recipient.
FR-005	Functional	The system shall provide a dashboard for Donor Partners to track the lifecycle status of their posted batches (Listed, Reserved, Picked Up, Expired).	Low	Pass: Dashboard reflects real-time status transitions corresponding to platform activity. Fail: Status on the dashboard lags or contradicts the active system state.	Provides necessary operational visibility and reporting metrics for the donor's CSR initiatives.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	The platform shall dispatch push notifications to verified shelters within a 5 km radius within 2.0 seconds of a high-quantity perishable food listing being posted.	High	Pass: Benchmarking confirms 95th percentile notification latency is ≤ 2.0 s under simulated peak load. Fail: Notification delivery exceeds the latency threshold.	Guarantees rapid mobilization of resources for highly perishable, large-volume donations.
NFR-002	Security & Compliance	The system must enforce mandatory OAuth 2.0 authentication and strictly verify non-profit organizational credentials before granting account activation to	High	Pass: Unverified accounts are blocked from accessing the active claim board. Fail: Public or unauthenticated users can reserve food batches.	Prevents fraud and ensures surplus food is strictly distributed to authorized charitable entities.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
		Shelter Coordinators.			

UML Diagram



Use-Case Flow Specification

Use Case ID: UC-05

Use Case Name: Reserve Food Batch

Primary Actor: Shelter Coordinator

Secondary Actor: Donor Partner (System Notified)

- **Preconditions:**
 1. The Shelter Coordinator is authenticated and has a verified non-profit status.
 2. The Donor Partner has an active, unexpired food batch listed on the platform.
 3. The Shelter Coordinator is within the permissible geographic radius to view the batch.
- **Postconditions:**
 1. The selected food batch is removed from the public active claim board and marked as "Reserved".
 2. A unique pickup verification code is generated and tied to the transaction.
 3. The Donor Partner is notified of the reservation and the selected pickup window.
- **Main Success Scenario (MSS):**
 1. Shelter Coordinator browses the active claim board and selects a specific surplus food batch.
 2. System displays batch details, including dietary tags, quantity, location, and the expiry countdown timer.

3. Shelter Coordinator selects Reserve Batch.
 4. System prompts the Shelter Coordinator to select an available pickup window (<<include>> UC6).
 5. Shelter Coordinator chooses a time slot and confirms the reservation.
 6. System verifies the batch has not expired and is still available in the inventory.
 7. System locks the batch, changes its status to "Reserved", and generates a secure verification code (<<include>> UC7).
 8. System displays the verification code and pickup instructions to the Shelter Coordinator.
 9. System automatically notifies the Donor Partner of the reservation and upcoming pickup schedule.
- Alternate Flow (A1 - Batch Expires or Claimed During Selection):
 - 6a. System detects that the expiry countdown timer has reached zero *OR* another shelter has concurrently reserved the final quantity of the batch during step 4/5.
 - 6b. System aborts the reservation lock.
 - 6c. System displays an error message: "This food batch is no longer available or has expired."
 - 6d. System automatically redirects the Shelter Coordinator back to the refreshed active claim board.